

EX.NO : 06

DATE :

DESIGNING RING AND MESH TOPOLOGIES USING CISCO PACKET TRACER

AIM :

To Designing a Ring and Mesh topologies by using Cisco Packet Tracer.

REQUIREMENTS :

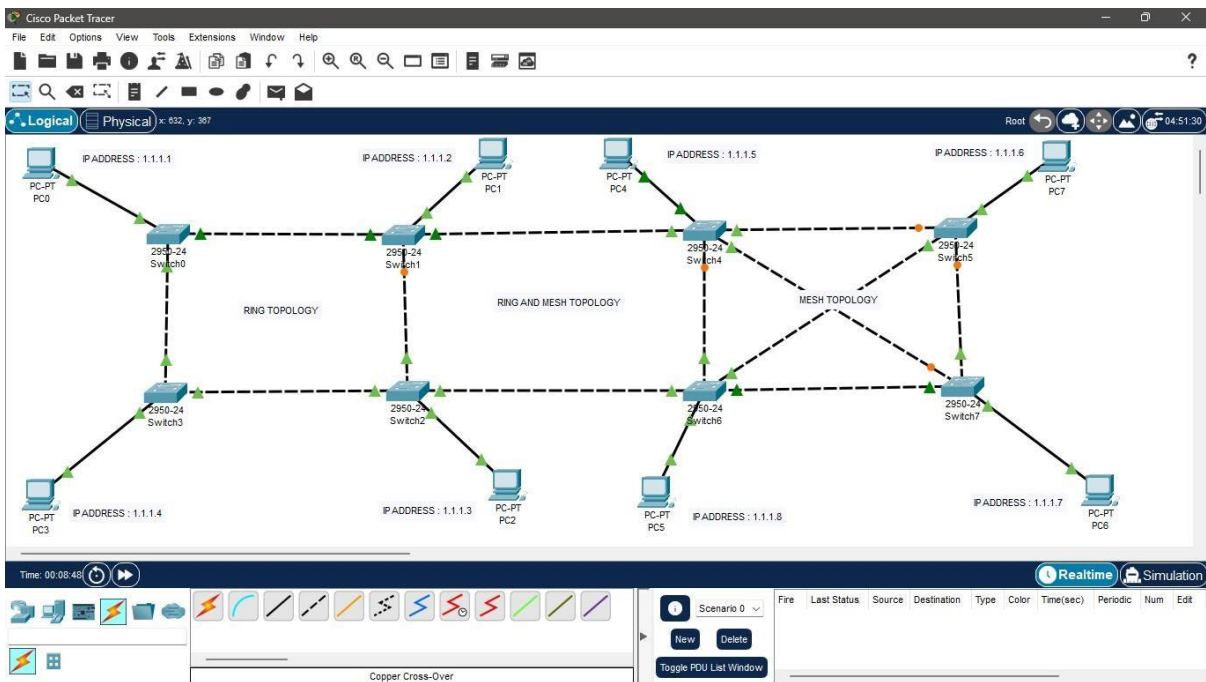
- 8 windows PC.
- 8 Switch (2950-24).
- 8 Straight Line LAN Cables.
- 12 Cross Over Cables.
- Cisco Packet Tracer.

PROCEDURES :

- Open CISCO PACKET TRACER software.
- Draw the 8 PC using END Device Icons.
- Draw the 8 Cisco 2950-24 Switch Using Switch icon lists.
- Make the Connections using Copper-Straight-Through Ethernet Cables.
- Make the Connections between Switches using Cross Overs Cables.
- Enter the IP Address To Each Machine.
- Check the Network Connections using Add Simple PDU(P).

NETWORK TOPOLOGY

RING AND MESH TOPOLOGY



HOST PC IP ADDRESS:

PC0 IP CONFIGURATION

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 1.1.1.1

Subnet Mask: 255.0.0.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

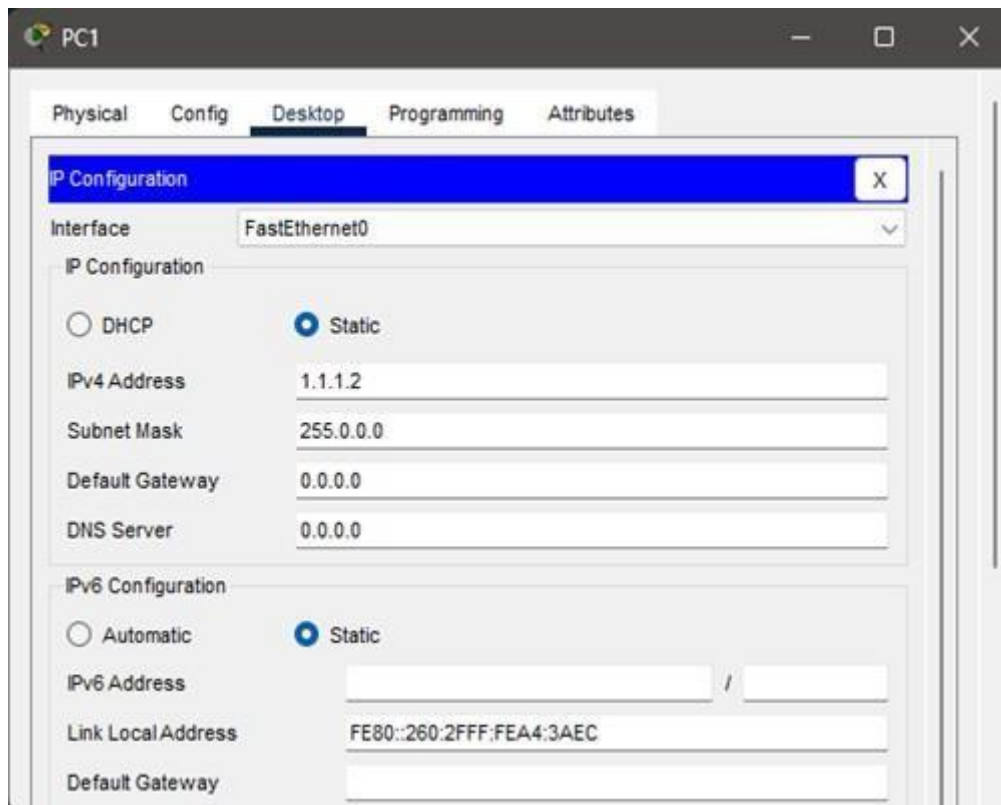
☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::210:11FF:FEC3:70B3

Default Gateway:

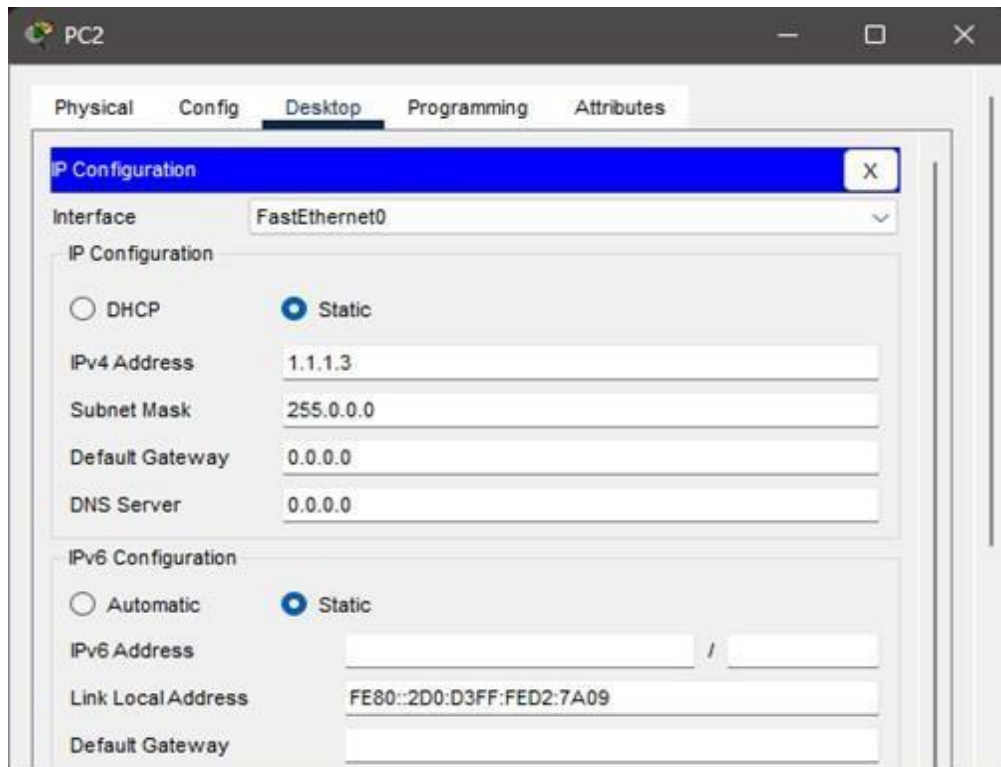
PC1 IP CONFIGURATION



The image shows the 'PC1' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The IPv4 settings are: IPv4 Address: 1.1.1.2, Subnet Mask: 255.0.0.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', 'Static' is selected. The IPv6 settings are: IPv6 Address: (empty), Link Local Address: FE80::260:2FFF:FEA4:3AEC, and Default Gateway: (empty).

Section	Option	Value
IP Configuration	Interface	FastEthernet0
	DHCP	☐
	Static	☒
	IPv4 Address	1.1.1.2
	Subnet Mask	255.0.0.0
IPv6 Configuration	Automatic	☐
	Static	☒
	IPv6 Address	
	Link Local Address	FE80::260:2FFF:FEA4:3AEC
	Default Gateway	

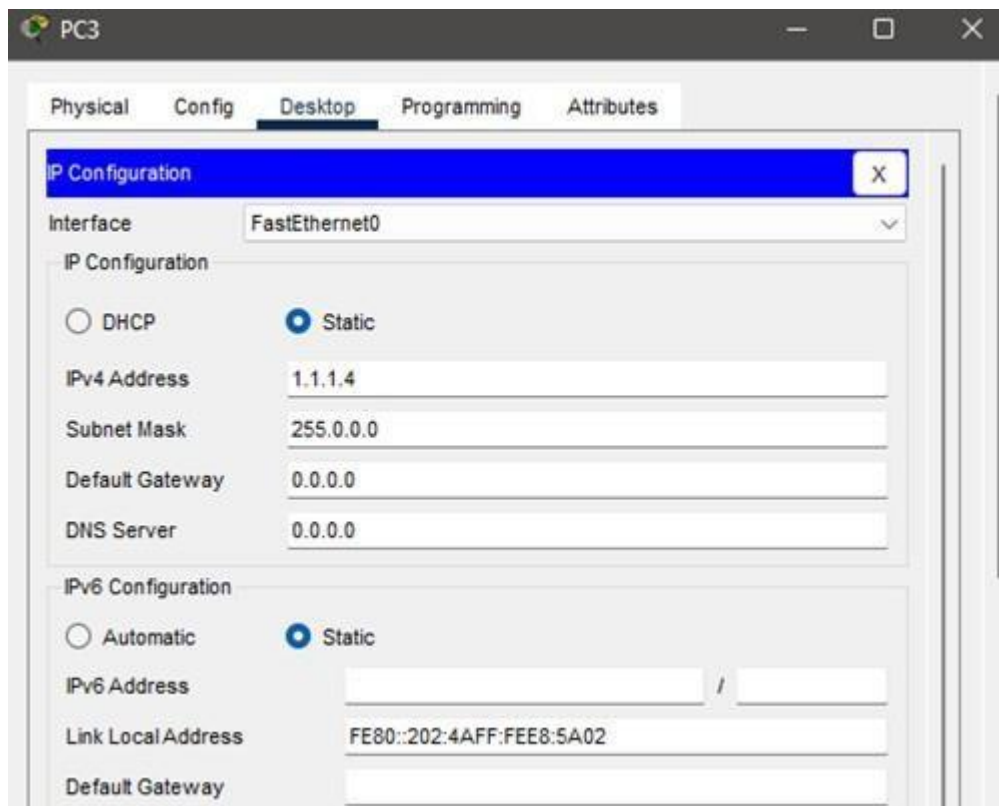
PC2 IP CONFIGURATION



The image shows the 'PC2' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The IPv4 settings are: IPv4 Address: 1.1.1.3, Subnet Mask: 255.0.0.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', 'Static' is selected. The IPv6 settings are: IPv6 Address: (empty), Link Local Address: FE80::2D0:D3FF:FED2:7A09, and Default Gateway: (empty).

Section	Option	Value
IP Configuration	Interface	FastEthernet0
	DHCP	☐
	Static	☒
	IPv4 Address	1.1.1.3
	Subnet Mask	255.0.0.0
IPv6 Configuration	Automatic	☐
	Static	☒
	IPv6 Address	
	Link Local Address	FE80::2D0:D3FF:FED2:7A09
	Default Gateway	

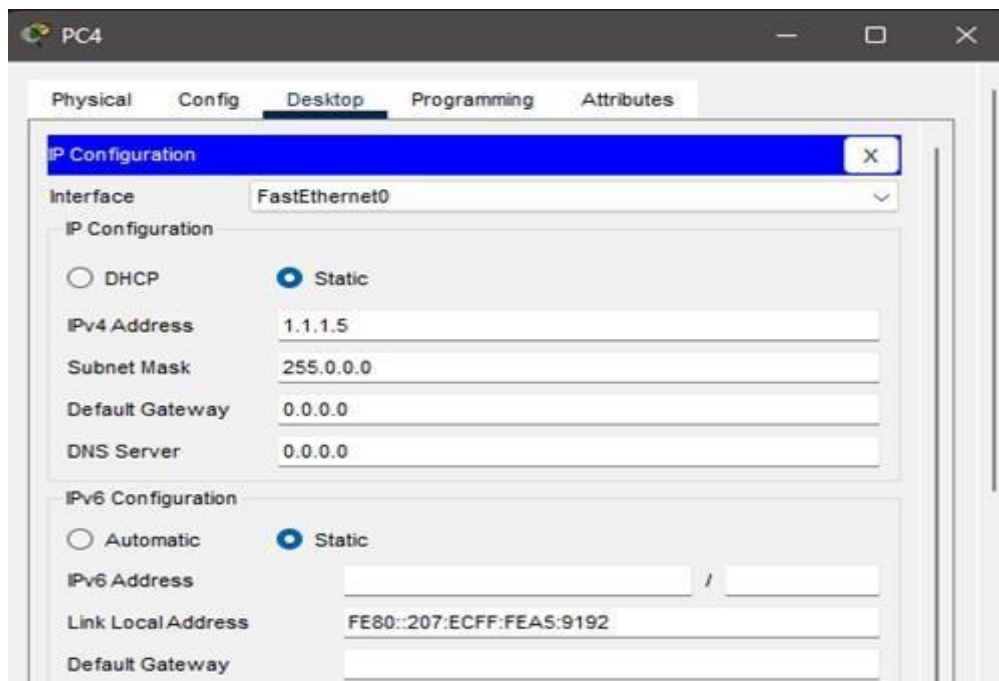
PC3 IP CONFIGURATION



The screenshot shows the 'IP Configuration' window for PC3. The 'Desktop' tab is selected. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The IPv4 settings are: IPv4 Address: 1.1.1.4, Subnet Mask: 255.0.0.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', 'Static' is selected. The IPv6 settings are: IPv6 Address: (empty), Link Local Address: FE80::202:4AFF:FEE8:5A02, and Default Gateway: (empty).

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.4
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::202:4AFF:FEE8:5A02
Default Gateway	

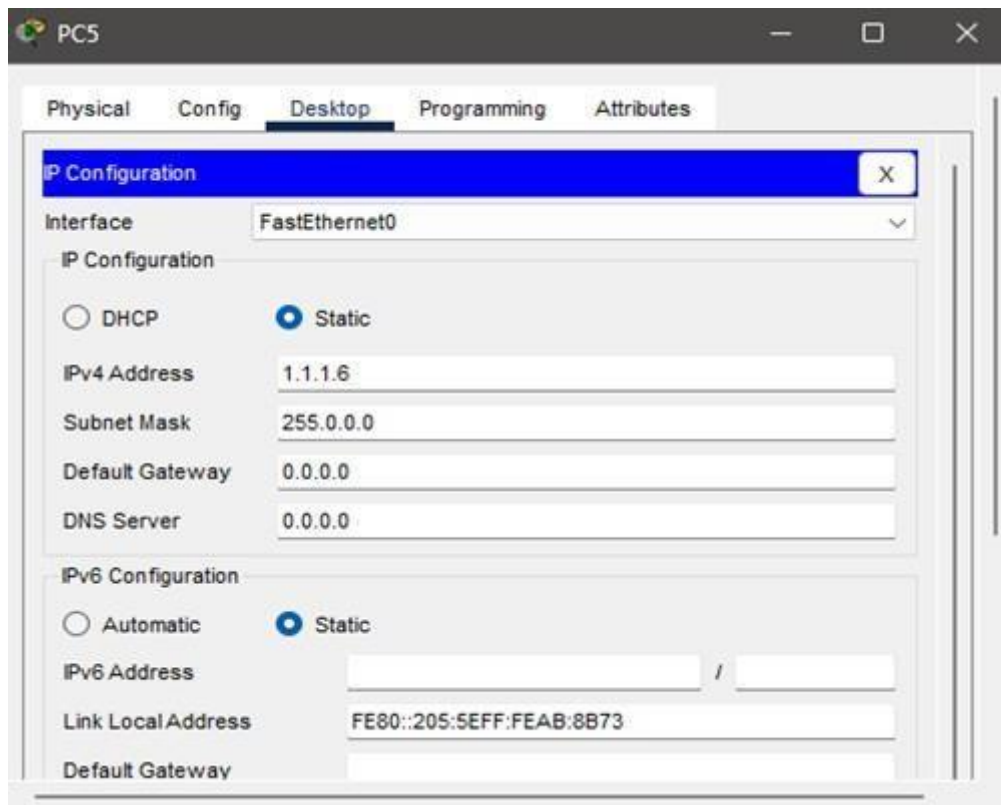
PC4 IP CONFIGURATION



The screenshot shows the 'IP Configuration' window for PC4. The 'Desktop' tab is selected. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The IPv4 settings are: IPv4 Address: 1.1.1.5, Subnet Mask: 255.0.0.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', 'Static' is selected. The IPv6 settings are: IPv6 Address: (empty), Link Local Address: FE80::207:ECFF:FEA5:9192, and Default Gateway: (empty).

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.5
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::207:ECFF:FEA5:9192
Default Gateway	

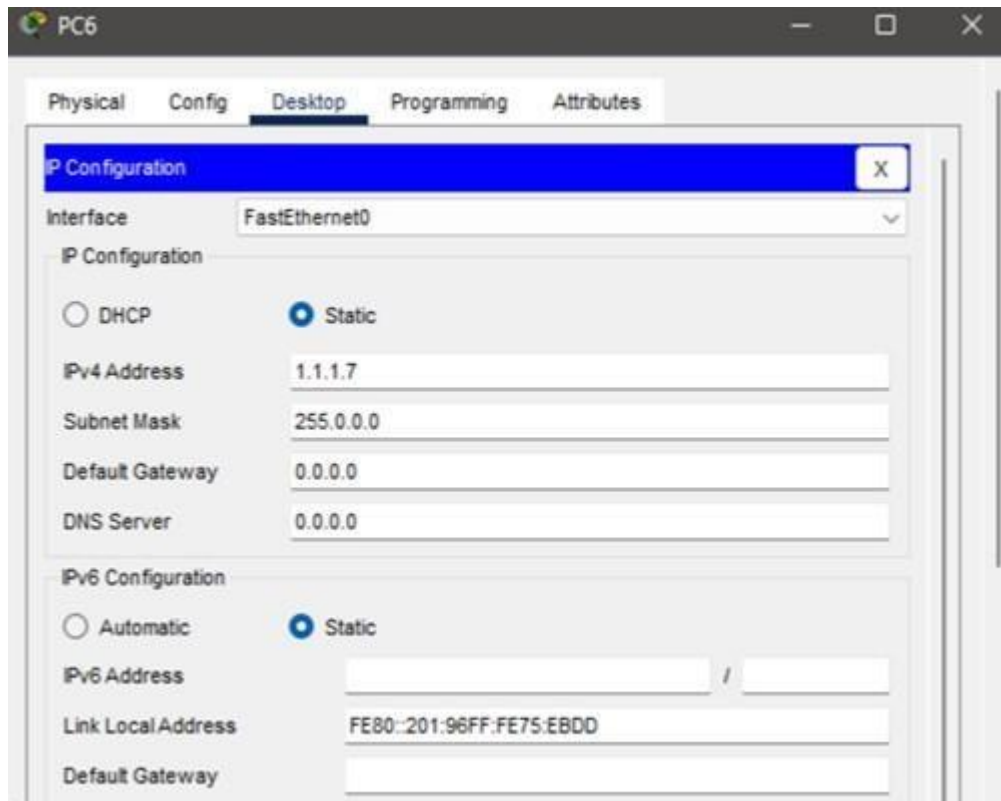
PC5 IP CONFIGURATION



The image shows the 'PC5' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The IPv4 settings are: IPv4 Address: 1.1.1.6, Subnet Mask: 255.0.0.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', 'Static' is selected. The IPv6 settings are: IPv6 Address: (empty), Link Local Address: FE80::205:5EFF:FEAB:8B73, and Default Gateway: (empty).

Section	Field	Value
IP Configuration	Interface	FastEthernet0
	Configuration Type	Static
	IPv4 Address	1.1.1.6
	Subnet Mask	255.0.0.0
	Default Gateway	0.0.0.0
IPv6 Configuration	Configuration Type	Static
	IPv6 Address	
	Link Local Address	FE80::205:5EFF:FEAB:8B73
	Default Gateway	

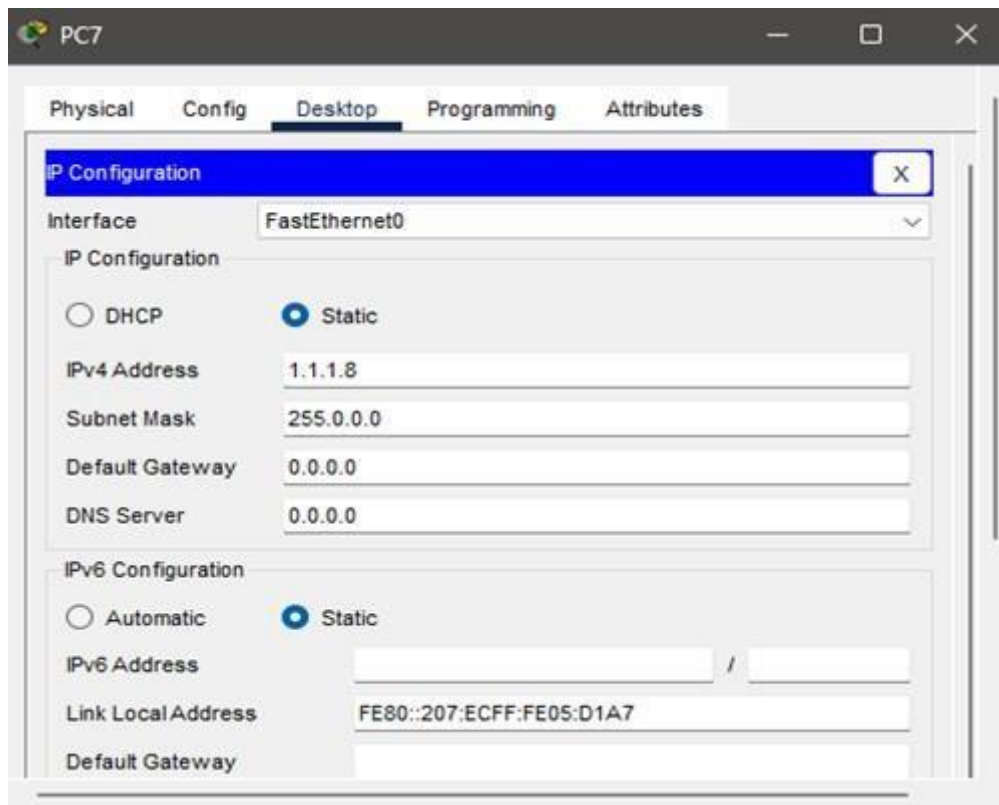
PC6 IP CONFIGURATION



The image shows the 'PC6' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The IPv4 settings are: IPv4 Address: 1.1.1.7, Subnet Mask: 255.0.0.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', 'Static' is selected. The IPv6 settings are: IPv6 Address: (empty), Link Local Address: FE80::201:96FF:FE75:EBDD, and Default Gateway: (empty).

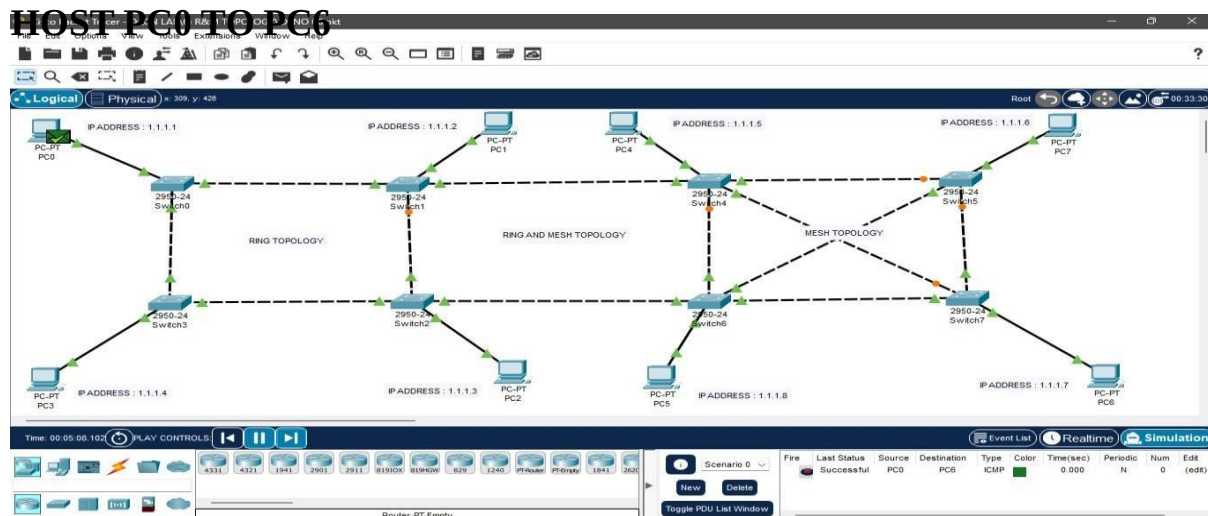
Section	Field	Value
IP Configuration	Interface	FastEthernet0
	Configuration Type	Static
	IPv4 Address	1.1.1.7
	Subnet Mask	255.0.0.0
	Default Gateway	0.0.0.0
IPv6 Configuration	Configuration Type	Static
	IPv6 Address	
	Link Local Address	FE80::201:96FF:FE75:EBDD
	Default Gateway	

PC7 IP CONFIGURATION

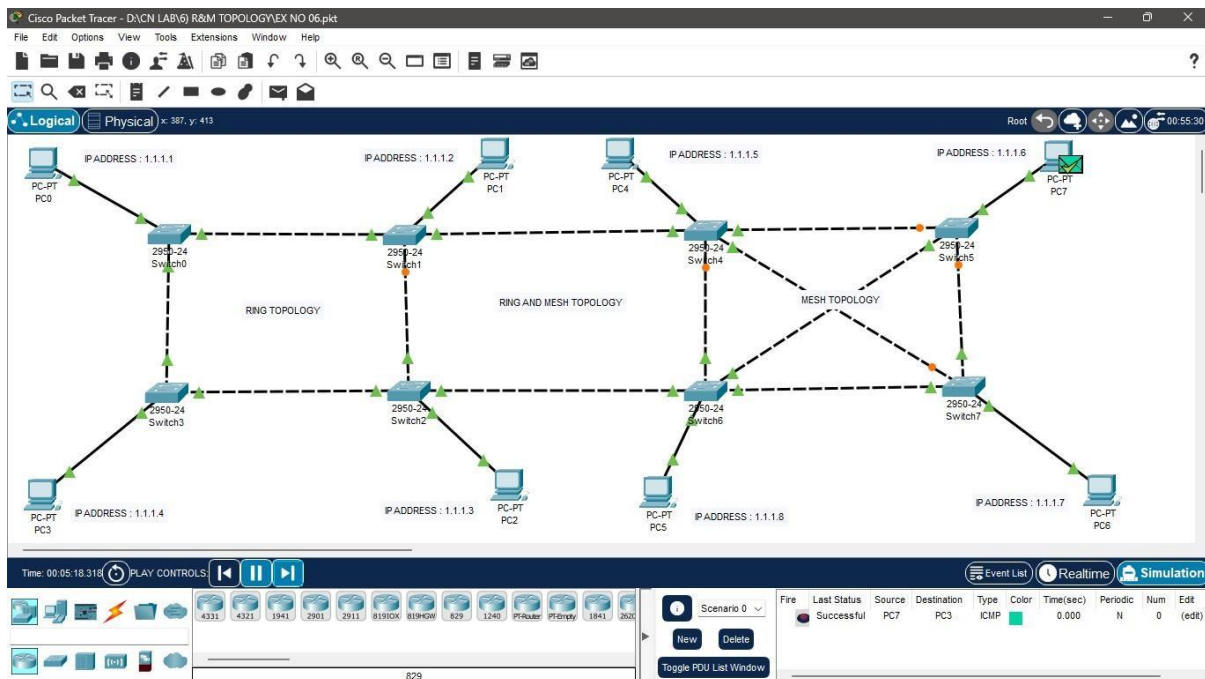


VERIFY LAN NETWORK CONNECTIVITY:

Using Add Simple PDU(p), Click the mail icon and then drop one mail to one of the PC in first lan and another mail to PC in another lan. If the resultant window shows the successful delivery, then network connectivity is successful.



HOST PC7 TO PC3



RESULT :

Thus, Ring and Mesh topologies are designed using cisco packet tracer and the communication between Ring and Mesh topologies is checked successfully.

